Aston University
BIRMINGHAM UK

Assessment Brief
AM41RS Specialist Research Skills and Techniques

Purpose of an assessment brief

(How should I use this brief to help with my assessment?)

This brief gives you key information and guidance about your assessment. It's the starting point for your work, and your module tutors will refer to it during assessment discussions. Read the brief carefully, along with any other materials your tutors provide, and use them to help you plan and complete your assessment.

1.0 Essential Information

Module Code and Title	AM41RSA Specialist Research Skills and Techniques
Module/Assessment Lead contact	Name and xxx@atson.ac.uk Ulysses Bernardet <u.bernardet@aston.ac.uk>
Assessment Title	Portfolio-based assessment
Assessment weighting	1. Peer feedback on presentations (30%) 2. Self-paced lab worksheets using the R software (Pass/Fail) 3. Dissertation proposal (70%)
Submission Date and Time	

Date	Time	Assignment
Tuesday 14 October	16:00	Presentation 1
Tuesday 21 October	16:00	Presentation 2
Wednesday 22 October	16:00	Peer feedback 1
Tuesday 28 October	16:00	Presentation 3
Wednesday 29 October	16:00	Peer feedback 2
Tuesday 4 November	16:00	Presentation 4
Wednesday 5 November	16:00	Peer feedback 3
Wednesday 12 November	16:00	Peer feedback 4
Thursday 13 November	16:00	Lab 1
Thursday 20 November	16:00	Lab 2
Thursday 27 November	16:00	Lab 3
Thursday 4 December	16:00	Lab 4
Thursday 11 December	16:00	Lab 5
Monday 5 January	16:00	Dissertation proposal

Word Count/Page Count/Lengths	Peer feedback: 100-150 words Dissertation proposal: 6 pages (excluding references)

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2.0 Assessment Overview

Study proposal and peer-review

For weeks 2,3,4, and 5 students are asked to prepare a short presentation (5-10min, 1-2 slides) on a topic specified at the end of the previous week's class. These assignments include, but are not limited to, designing, and executing small mock-up studies. In weeks 2,3,4, and 5 some students are randomly selected to present their work. After class, all students must submit two peer feedbacks of **two** presentations **that were not discussed in class**. This **peer feedback is the graded assignment.** The feedback will not be disclosed to the author of the presentations.
The peer feedback of **100-150 words** should

- highlights both strengths and weaknesses of the presented methods (what and why)
- suggests how to improve the methods

Submission:
In weeks 2,3,4, and 5 **each student** has to submit on Blackboard the following **two items:**

- **Two days before class by 16:00:** Presentation on the assigned topic In **PDF** format.
- By the **Wednesday** following class **16:00:** Peer feedback on **two** presentations

Note that a student cannot write the feedback about their own presentation.

Lab worksheets

In weeks 6, 7, 8, 9, and 10 each student has to submit the lab worksheets by **16:00** on the day of the lab. The idea is that these worksheets are completed **during** the lab sessions.

Dissertation proposal

For this assignment, students are expected to write a detailed project proposal which describes the aims of the project and the project deliverables. The topic can be chosen from the list provided or student proposed, subject to approval.

The document containing a detailed project proposal which describes the aims of the project, and the project deliverables should NOT exceed **6** pages (excluding References and Appendices).

The proposal **must be typeset in LaTeX**. A template is provided on Blackboard.

Your project proposal should include the following five components:

a. **Potential benefits.** The motivation behind the project should be described along with the potential benefits of the project. Benefits in four categories are envisaged:
    - **Academic**: The project will contribute to our understanding of this problem domain in some way.

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- **Technical**: The project will identify and/or evaluate a new way of doing something, deliver a prototype system that explores new ideas, or deliver a solution to an identified problem.
- **Business**: Here a business case should identify any commercial partner associated with the project and fully describe the financial or other benefits of the project to that partner.
- **Social and Environmental**: The project will have potential social or environmental benefits.

Note that projects do not have to have benefits under all four headings listed above but the proposal should highlight potential benefits under at least one heading.

b. **The main problems or research questions to be addressed, the chosen methodology and its ethical implication**.
   - The proposal should identify the main problems to be solved and describe how those problems will be addressed in your MSc project.
   - You should give careful attention to risk assessment (i.e. what happens if a problem is harder than expected) and should outline your proposed project management.
   - You should outline the methodology to be adopted for this project.
   - You should also discuss any potential wider ethical issue that the work to be carried out may have. The "Ethics and Information Technology" journal published by Springer is a good journal that may help you when considering the ethical implications of your project.

   Note that you are not expected to complete an application for ethical approval before submitting your dissertation proposal. You are only expected to give consideration to any ethical implications at this stage.

c. **Requirements and feasibility.** There may not have been time in the initial stage to fully detail the requirements. However, you should have at least an outline of the full requirements. You should assess the feasibility of the project proposal and provide an argument that what you are proposing will meet the requirements within the available time scale.

d. Project Plan. You should provide a project plan identifying the tasks to be completed and the schedule for completing those tasks. The project plan should be presented as a Gantt chart or other suitable diagrams. The project plan should detail the activities for the whole duration of the project period. The activities in the project plan should be explained and justified.

References. The list of References has no prescribed limit.

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Submission Details:

- All assignments require submissions on Blackboard by the specified deadline dates.
- Links to submissions are found in the “Assessment Submission” content area on Blackboard.
- The weekly presentations have to be uploaded as a pdf and be named as follows: <first name>_<last name>-Week<#>.pdf
- The Dissertation proposal has to be uploaded as a pdf and be named as follows: <first name>_<last name>-Proposal.pdf
- Lateness penalty is 10% per calendar day.

There is no explicit penalty for exceeding the length limits of the submissions (200 for the commentaries/peer feedback, 6 pages for the Dissertation proposal), however, **content beyond the limit will be disregarded**

3.0 Purpose of Assessment

This module aims to ensure that students develop their research and professional skills at maximum, including critical thinking, experimental methodology and literature search. An important aim is that students gain practical experience in effective communication on a variety of topics within DS/AI to a wide range of audiences and taking different formats, therefore the portfolio of assessment will include a written report, peer evaluation of an oral presentation, and practical statistics exercises:

- MSc project proposal, which is a plan on a chosen topic (70%), assessing LO1, LO3 and LO4. This will give students an early opportunity to investigate topics for dissertation.
- 5 self-paced lab worksheets using the statistics software (SPSS or R, for example, all pass/fail), assessing LO2. This will give students the opportunity to demonstrate their application of practical stats skill on descriptive and inferential statistical analysis.
- Peer evaluation of oral presentation on a given experimental study topic, recorded and peer assessed (30%), assessing LO1. This will give students the opportunity to practically apply the theoretical material presented in the lectures and to train their ability to provide constructive criticism. The assessment itself is the written critical peer evaluation.

For clarity on the oral presentation:
Every week the student will prepare a short presentation about a proposed study for the following week (designing and executing small mock-up studies, 3-4 slides). Some students will be selected randomly to present their work, and every student has to write 2 short review comments (100-150 words) on other students' presentations (made available via Blackboard). Failure to submit a presentation on the assigned topic will incur a penalty of TWO points.

4.0 Feedback Expectations

Presentation feedback will be in the form of a breakdown of marks using a marking

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rubric and a summary of comments raised during no-right-of-reply session. These will provide formative feedback to the student for the report.
Report feedback will be in the form of a mark and a short summary.
Feedback on execution of the stats lab worksheets will be throughout lab sessions.
Feedback on progress with understanding of research methods will be provided throughout the first half of the module, in the live sessions and in the form of peer feedback.
Feedback on writing skills will be provided via interactive practical writing sessions.
Feedback on critical writing will be provided via the assessment of peer evaluation.
The oral presentations will be recorded, and the recording made available to students for peer- and self-assessment.

5.0 Guidance on AI Usage

Can I use AI in my work and how?

Use of Gen AI is Optional	The dissertation proposal needs to be individual and original, written by you. **Use of ChatGPT or other Large Language Models for the writing of the text of the proposal is not acceptable.** It is appropriate, but not essential, to make **constructive use of generative AI tools** as described below. This should not include using AI to create the content of the assessment submission, but may include: 1) Generating initial ideas 2) Finding references for a starting point 3) Understanding of papers. Any application or use of generative AI must always be acknowledged. Misuse of generative AI, as defined in Aston's assessment regulations, will be treated as an academic offence and may be subject to disciplinary action.

6.0 Assessment Criteria and Rubrics

Note: **All elements of assessments have to be passed to pass the module**

Peer-Feedback

Each of the two feedback will be graded in the range 0-3:
0 Failure to submit
1 Poor feedback that does not provide useful information to the author.
2 Good feedback that is sensible but does not include novel arguments.
3 Feedback that is presented in a critical, yet constructive way. The feedback includes aspects that were not raised in class.
The final mark of the peer feedback is the average of the best three (out of four)

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submissions and will account for 30% of the overall mark.

Important: **Failure to submit a presentation on the assigned topic, will incur a penalty of TWO points.**

Lab worksheets

The lab worksheets are graded as PASS (100) or FAIL (0). Any reasonable submission will receive a pass and no formative feedback is provided. **To pass this assessment element, at least 3 or the 5 labs need to be passed.**

Dissertation proposal

0-39 There is a minimal attempt. It is not clear what the dissertation work will be and/or how it would be placed in context. Essential elements, such as objectives, ethical implications, the methodology or the plan are missing or too generic.

40-49 There is a meaningful attempt, however elements such as the objectives, the methodology or the plan may be missing, the writing is not of suitable standard for communication. Ethical implications of the project are not considered sufficiently.

50 Proposal is acceptable, although limited in scope. The components of the project plan are all present but will be superficial or brief. Appropriate consideration was given to the ethical implications of the project.

51-59 The project work is placed in context. The work is motivated in general, and a range of specific objectives is set. Any experimental objectives of the work are stated with suitable motivation. The potential benefits and the main problems to be addressed have been identified, but the proposed methodology contains issue(s) that will need addressing during the project development. A project plan that needs major modification, either because it is over-ambitious, not challenging enough or lacking in detail may indicate that the student does not have a sufficient understanding of the project topic. Appropriate consideration has been made on the ethical implications of the project.

60-69 The potential benefits and the main problems/research questions to be addressed have been discussed in the proposal. The proposed methodology is appropriate. The identified project requirements are appropriate and clearly presented. The schedule given in the project plan is achievable within the time scale without major modification. Appropriate consideration has been made on the ethical implications of the proposed work.

70-79 There is evidence of both thoroughness (e.g. in attention to detail and scope) and depth of insight into the problems raised by the work and the range of objectives that the work should pursue. Project plan is well thought through and justified. Good consideration has been made on the ethical implications of the proposed work.

80-89 The proposed work is placed in the context of critical review of existing processes within the domain (or the business, if applicable), preferably coupled with suggestions for process improvement or new processes. Thorough consideration has been made on the ethical implications of the proposed work. Project plan is of

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professional standard. 90+ There is some original insight in the arguments for the motivation of the project work. The work will clearly be at the forefront of current knowledge and/or professional practice.

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